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Integrated Vibration Analysis System for Enhanced Understanding of Drilling Dynamics and Optimization of Drilling Efficiency

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Abstract

This study presents the development and field validation of an Intelligent Saver Sub (ISS) designed to enhance the understanding of drilling dynamics and optimize drilling efficiency. The ISS is equipped with an integrated vibration measurement system capable of high-frequency, real-time acquisition, and wireless transmission of triaxial acceleration and rotational speed data at the top of the drill string. This system significantly improves data acquisition precision and system response speed. Field tests demonstrate the ISS's stability, high data integrity, and accurate anomaly detection, highlighting its potential for practical engineering applications. The introduction of a Bit Vibration Bounce Index (BVI) further enhances on-site intervention efficiency.

Introduction

As a critical strategic resource and pillar of economic development, the exploration and exploitation of oil and gas resources heavily rely on technological breakthroughs in drilling engineering. Among these, the dynamic behavior of the drill string directly determines the quality of wellbore-reservoir connectivity and operational safety. Constrained by heterogeneous formations, the drill string system is subjected to combined axial-torsional-lateral dynamic loads, which can induce nonlinear vibration modes such as whirl and stick-slip. These phenomena significantly reduce drilling efficiency, intensify drill string fatigue damage, and may even lead to severe downhole tool failures. Therefore, vibration monitoring and analysis during drilling have become a key research focus in the field of drilling engineering.

Existing studies generally attribute downhole vibrations to nonlinear interactions between the bit, drill string, and formation [1,2], typically categorized into three modes: axial, lateral, and torsional vibrations, as shown in Figure 1. Axial vibration (bit bounce) refers to the periodic elongation and contraction of the drill string along its axial direction, primarily caused by an uneven bottomhole profile and uneven interaction forces between the bit cutters and the fractured rock. This phenomenon is particularly pronounced in vertical

wells through compact formations and when using roller cone bits. Severe bit bounce can generate strong impact loads, leading to cutter chipping or even drill string fracture. Torsional vibration (stick-slip vibration) originates from the frictional interaction between the rotating drill string and the wellbore wall, commonly observed in extended reach and deep well operations [3]. When the torsional strength of the drill string is insufficient to overcome the frictional resistance of the wellbore, a periodic "stick-slip" motion occurs. This type of vibration can easily cause drill string torsional failure, bit tooth breakage, and downhole tool malfunction. Lateral vibration manifests as the orbital revolution (whirl) of the bit and bottom hole assembly (BHA) around the wellbore axis. Contributing factors include bit-rock interaction, drill string–wellbore collision, eccentricity, and bending deformation. Prolonged whirl not only leads to unilateral wear of the BHA, bit damage, and wellbore enlargement, but may also trigger wellbore instability. Moreover, repeated lateral vibrations accelerate drill string fatigue damage and significantly shorten its service life [4]. The coupled effects of these three vibration modes further increase the risk of downhole tool failure. Therefore, in-depth investigation into the vibration characteristics of drill strings is of great significance for ensuring drilling safety and efficiency.

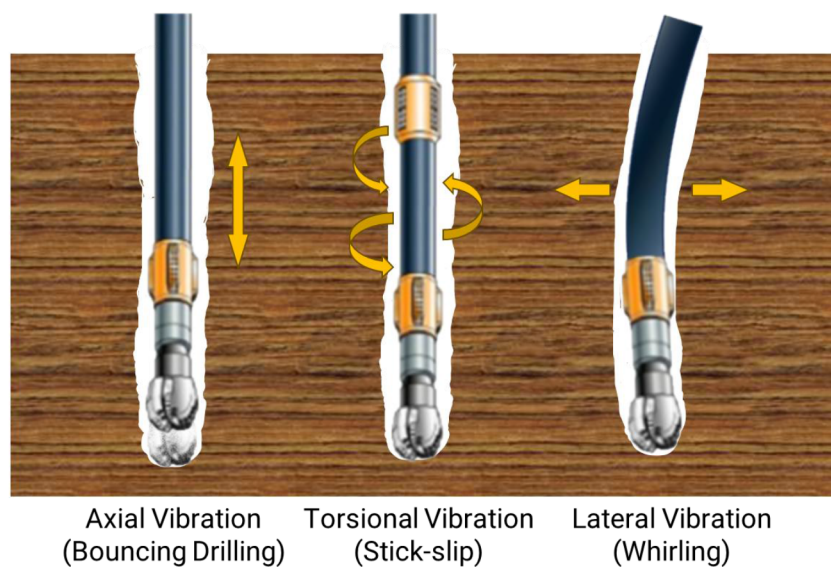


Figure 1—Schematic diagram of drill string vibration modes [1].

With the continuous advancement of sensing and measurement technologies, high-frequency measurement of drill string vibration frequencies using high-precision sensors has become a feasible approach [5]. Depending on the measurement location, current mainstream measurement methods can be broadly categorized into two types: downhole measurement and surface measurement. Downhole measurement involves installing sensors within the bottom hole assembly (BHA) to obtain accurate information regarding drill string vibrations [6–8]. However, such downhole tools suffer from limited real-time data transmission capabilities due to the bandwidth constraints of mud pulse telemetry, making real-time transmission of high-frequency vibration signals unachievable. As a result, memory-based measurement is more commonly employed. To address these limitations, researchers have shifted focus toward developing surface-based vibration measurement tools [9,10]. These tools offer minimal interference with drilling operations, convenient installation, and real-time data transmission capabilities. However, the accuracy of surface measurements is limited, particularly for lateral vibrations, which experience significant attenuation when transmitted to the surface.

To address the aforementioned limitations, we have developed an intelligent saver sub (ISS) equipped with an integrated vibration measurement system. This system enables real-time measurement of triaxial acceleration and rotational speed at the top of the drill string, overcoming the limitations of existing surface

data logging systems. The ISS is equipped with high-precision calibrated sensors capable of achieving a sampling frequency of 1500 Hz, with data wirelessly transmitted to surface computing and display devices. To avoid loss of vibration measurement accuracy in complex wells, we also employ a downhole vibration measurement (DVM) sub, which uses memory-based logging to collect downhole vibration data at a sampling frequency of 1500 Hz. In addition, a vibration monitoring software system has been developed to continuously track the measurement data from the ISS, enabling prompt detection of abnormal vibration patterns. This integrated vibration analysis system has been successfully implemented and validated in a development well in the field.

Integrated Vibration Analysis System

Intelligent Saver Sub

The intelligent saver sub (ISS) with an integrated vibration measurement system (as shown in [Figure 2](#)) is a surface vibration measurement tool specifically designed for installation between the top drive unit and the drill pipe (as illustrated in [Figure 3](#)). This device not only retains the mechanical protection functionality of a conventional saver sub but also integrates a high-precision vibration sensor system capable of acquiring high-frequency vibration data from the drill string under drilling conditions in real time. Through the built-in wireless data transmission module, the system can stably and reliably transfer the collected vibration signals to the vibration processing and analysis software platform located in the surface driller's cabin (as shown in [Figure 4](#)), thereby enabling continuous monitoring and dynamic analysis of the vibration status during drilling operations. The application of this technology provides data support and advisory for vibration control of the drill string system, optimization of drilling parameters, improvement of the rate of penetration, and extension of bit service life.



Figure 2—Intelligent saver sub.

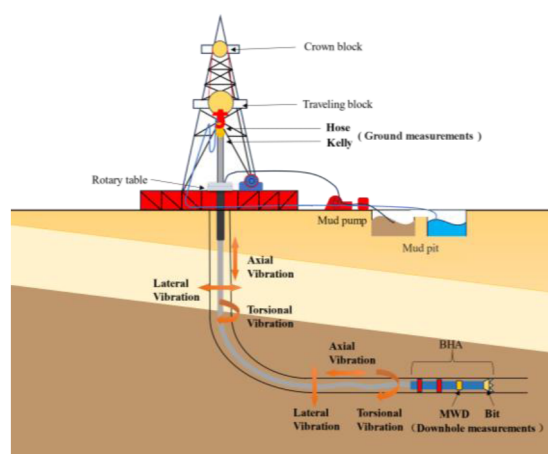


Figure 3—Installation positions of surface and downhole tools [1].

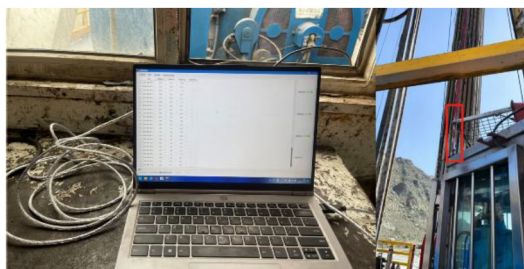


Figure 4—Wireless transmission status.

The ISS is capable of real-time measurement and transmission of triaxial amplitude data, drill string rotational speed, and environment temperature. It can operate continuously and stably for extended periods, enabling long-duration vibration monitoring and real-time signal transmission. To ensure the integrity and reliability of data acquisition, and to minimize potential data loss during wireless transmission, the system incorporates an internal data storage module with substantial capacity. This module allows for local storage of all raw data throughout the entire operational cycle, effectively guaranteeing data traceability and completeness.

Downhole Vibration Measurement

The downhole vibration measurement tool is a robust, highly integrated multi-parameter measurement device (as shown in Figure 5), specifically designed to withstand the complex and harsh operating conditions encountered during drilling operations. Installed at critical locations within the bottom hole assembly (BHA) (as illustrated in Figure 2), the tool is capable of synchronously acquiring and recording multiple engineering parameters, including triaxial vibration acceleration, rotational speed, temperature, torque, annular pressure (both inner and outer), and weight on bit. The sampling frequency of the vibration signals is high, while the data storage frequency is set to ensure efficient and high-resolution dynamic monitoring. The tool can continuously and stably operate for more than 200 hours under high-temperature conditions of up to 150 °C. It is equipped with an internal storage capacity sufficient to meet the long-term evaluation requirements for downhole conditions and the mechanical performance of drilling tools under complex operational scenarios. In this study, the data acquired by this tool were used to calibrate and validate the measurements obtained by ISS, thereby enhancing the overall reliability and engineering applicability of the monitoring system.



Figure 5—Downhole vibration measurement.

Intelligent Monitoring and Analysis System for Drilling Vibration

The intelligent drilling vibration monitoring and analysis system can real-time receive key engineering parameters—such as vibration, rotational speed, and temperature—acquired by ISS. It supports low-latency transmission of high-frequency data, ensuring the timeliness and completeness of downhole dynamic information. The system features stable and reliable communication capabilities, enabling it to meet data transmission requirements under complex drilling operational conditions. To enhance data transmission integrity and system fault tolerance, a data caching mechanism has been specifically implemented within the ISS. This mechanism allows for resumption of data transmission from breakpoints and retransmission of lost

data in the event of communication interruptions, effectively preventing data loss and ensuring the continuity and accuracy of the monitoring data. The overall system operates stably, demonstrating strong engineering applicability and anti-interference capability, making it suitable for long-duration, high-intensity drilling field applications.

System Technical Architecture. The system's technical architecture consists of three core layers: the presentation layer, application layer, and data layer, as shown in Figure 6. The overall structural design is rational, with clear functional partitioning, and exhibits good modularity and scalability. The system utilizes widely adopted programming languages for the application and data layers, leveraging their mature ecosystems, high stability, and strong cross-platform capabilities in enterprise-level application development.

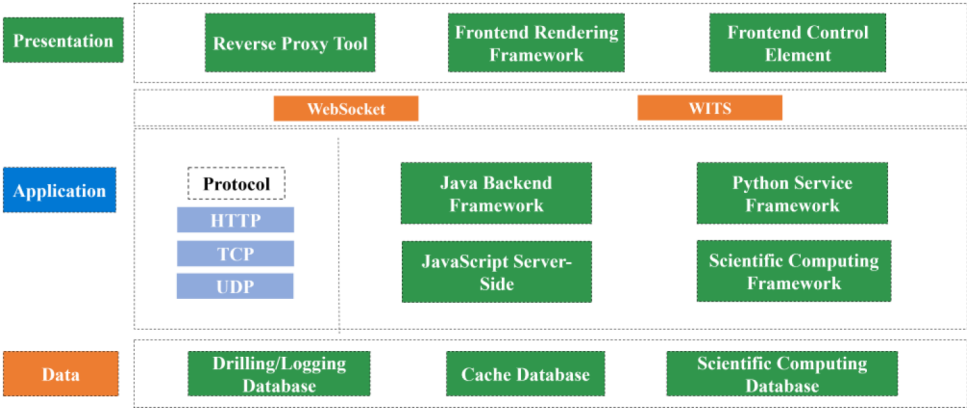


Figure 6—System technical architecture.

The system supports mainstream Internet of Things (IoT) communication protocols and has achieved seamless integration with ISS, various sensors, and widely used logging instruments, demonstrating excellent compatibility and integration. To address the diverse data storage and access requirements of different functional modules, the system employs multiple database types for categorized management, meeting the demands of high efficiency, high reliability, and high scalability. The front-end interface integrates a rich set of charting and graphical components, supporting multi-dimensional data visualization, which enhances user interaction experience and improves the intuitiveness and diversity of information representation.

System Modules. The system is composed of multiple functional modules working in coordination, including the user management module, hardware and software data source configuration module, acquisition task configuration module, real-time monitoring module for high-frequency drilling vibration data and logging data, as well as the historical data query and analysis module. These modules have clearly defined functions and well-structured interfaces, collectively enabling comprehensive monitoring of downhole vibration conditions, data acquisition, real-time analysis, and historical data traceability. This modular design effectively supports dynamic analysis and decision-making requirements during drilling operations.

Experimental Setup and Data Analysis

Experimental Setup

To verify the effectiveness of the Integrated Vibration Analysis System in actual drilling operations, a field test was conducted. A horizontal well was selected as the test well, with the objective of evaluating the

system's vibration monitoring capability and data acquisition performance during drilling under complex geological conditions.

The horizontal well had a designed total depth of 2080 meters and a vertical depth of 1252 meters. It featured a typical complex wellbore structure, with specific structural parameters detailed in [Table 1](#). By deploying the integrated vibration analysis system during the drilling process, vibration response data under key operating conditions were successfully acquired, providing reliable data support for subsequent system performance evaluation and drilling process optimization.

Table 1—Wellbore structure of test well

Bit Run Sequence	Interval(m)	Bit Size(mm)	Bit Type
First Spud	0-120	Φ311.15	PDC
	120-1427	Φ215.19	PDC
Second Spud	1427-2080	Φ215.19	PDC

Data Acquisition Performance

During the field test, surface ISS and downhole DVM were deployed and operated simultaneously starting from the second spud-in on August 4, and continued until the completion of tripping operations on August 10, maintaining stable operation for a total of six consecutive days.

During this period, the ISS successfully achieved continuous acquisition and real-time transmission of key parameters such as vibration, rotational speed, and temperature throughout the drilling process. Meanwhile, the DVM synchronously recorded multiple engineering parameters, including tri-axial vibration, weight on bit, torque, and inner/outer annular pressure. The overall data acquisition integrity of the system was high, with the vibration data sampling frequency stably maintained, meeting the requirements for high-resolution dynamic monitoring.

The acquired raw data were not only transmitted in real time to the surface monitoring system via a wireless communication module, but also fully backed up using the ISS's built-in local storage unit, ensuring data security and traceability. The collected data covered a variety of typical operating conditions during the drilling process, including but not limited to normal drilling, making a connection, tripping in/out, and formation transitions. These data provide a rich and reliable foundation for subsequent analysis of drill string vibration characteristics, downhole condition identification, drilling parameter optimization, and the development of fault early warning models.

Data Analysis

Taking the acquired vibration data on August 9, 2024, as an example, a comparative analysis was conducted between the vibration data collected by surface ISS and downhole DVM. The results indicate a high degree of consistency in the overall vibration trends between the two systems, as shown in [Figure 7](#). The vibration variation patterns recorded by ISS and DVM are largely aligned during multiple key time intervals, confirming the reliability and accuracy of the ISS in dynamic response monitoring.

However, certain discrepancies still exist in terms of vibration amplitude. This deviation is consistent with findings from previous studies and is primarily attributed to differences in the downhole installation positions of the two devices, which result in varying distances from the vibration source. These positional differences influence the measured vibration intensity. In addition, factors such as sensor type, minor differences in sampling frequency, and the influence of local structures on vibration propagation may also contribute to the observed data discrepancies.

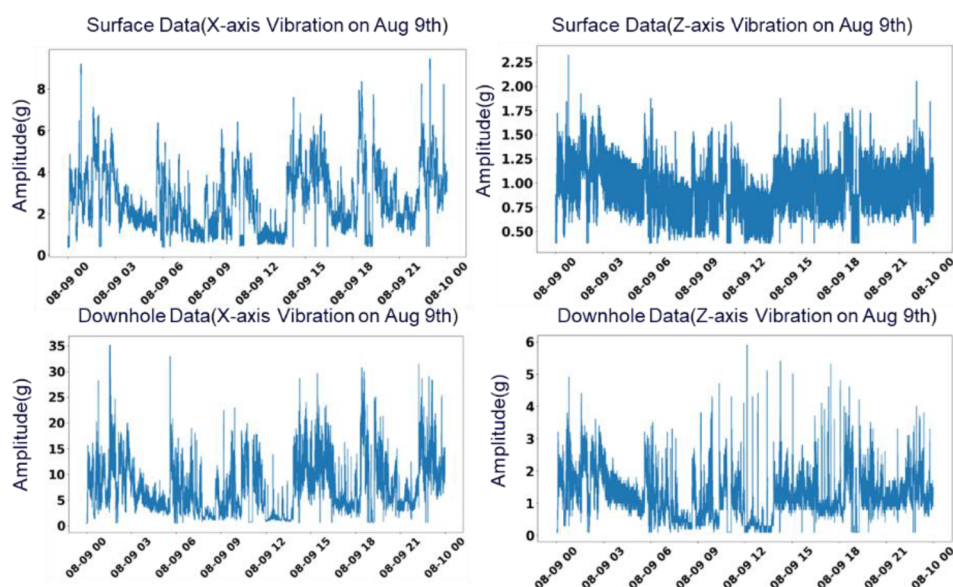


Figure 7—Comparison of ISS and DVM data.

In summary, despite the presence of certain numerical differences in amplitude, the high consistency in overall vibration trends between ISS and DVM demonstrates that the ISS possesses strong engineering applicability and is capable of providing effective support for vibration monitoring and analysis during drilling operations.

Vibration Monitoring and Bit Bounce Index Analysis Results

Through continuous monitoring of the longitudinal vibration data, the root mean square (RMS) value of the vibration signal in each second is extracted as the vibration intensity at that moment. This process constructs a time-vibration intensity sequence, as illustrated in Figures 8, which depicts the variation in longitudinal vibration data over a specific time. Based on this analysis, a Bit Bounce Vibration Index (BVI) is further calculated to quantify the severity of vibration anomalies and to enable automatic early warning functionality.

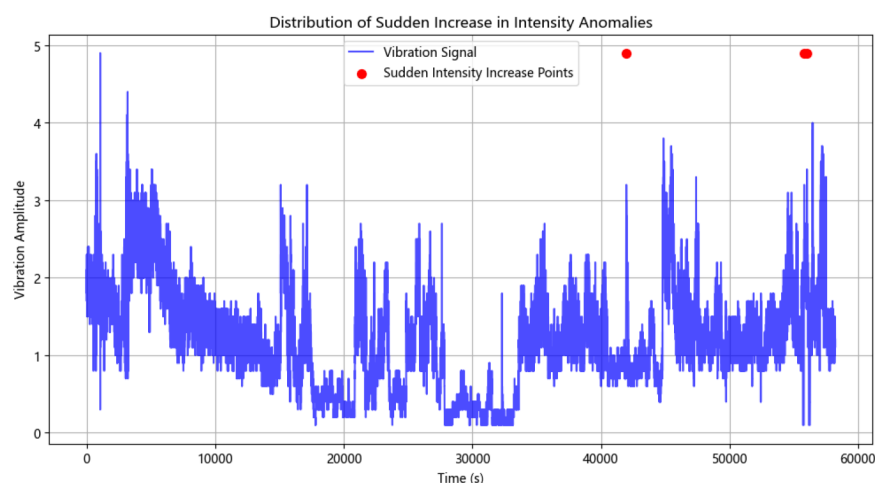


Figure 8—Distribution of sudden increase in intensity anomalies.

For example, at the timestamp of 56016 seconds, the system successfully detected a sudden increase in vibration intensity and promptly calculated the corresponding BVI in real time. When the index exceeds a predefined threshold (typically 0.6), the system triggers an alarm mechanism, alerting field operators to

monitor the current drilling conditions and take appropriate control measures to prevent potential equipment damage or downhole incidents, as shown in Figure 9.

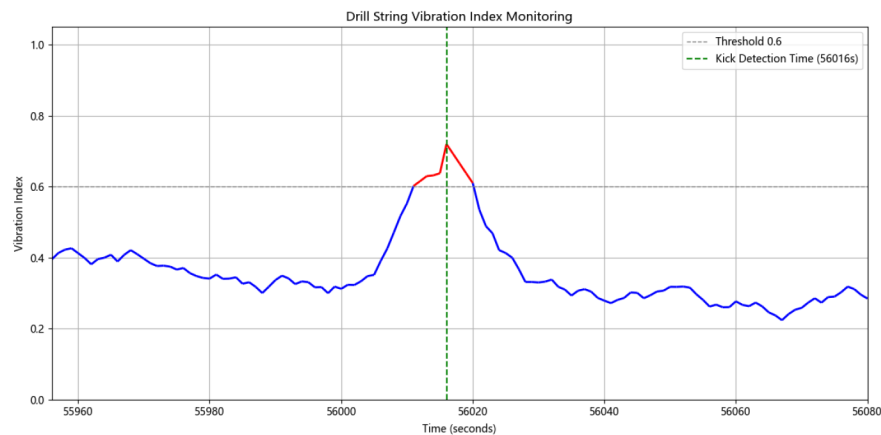


Figure 9—Bit bounce vibration index.

Using this index-based early warning mechanism, the system significantly enhances the real-time performance and accuracy of vibration anomaly detection in drilling operations, providing strong support for safe control and intelligent decision-making during drilling processes.

Conclusions

To address the challenges of low frequency and high latency in downhole vibration monitoring, this study developed an Intelligent Saver Sub (ISS) equipped with an integrated vibration measurement system. This system facilitates high-frequency real-time acquisition and wireless transmission of triaxial acceleration and rotational speed data at the top of the drill string, thereby significantly enhancing data acquisition precision and system response speed.

The ISS is synchronized with the Downhole Vibration Measurement (DVM), which records downhole vibration data at the same sampling rate. This synchronization was utilized to benchmark the surface measurements. The benchmark results confirm that the surface ISS can reliably record vibration signals of the drill string. Additionally, a Bit Vibration Bounce Index (BVI) has been introduced as a quantitative early-warning indicator to improve the efficiency of on-site interventions.

Field testing has demonstrated that the system operates stably, maintaining high data integrity and accurate anomaly detection, indicating strong potential for practical engineering applications. Future developments will extend the system to complex-structure wells, deep wells, and intelligent drilling operations, providing robust monitoring and advisory capabilities for safe and intelligent drilling practices.

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